

How the Police Solve a Crime

The Crime Scene Investigator

What is a crime scene investigator?

Crime Scene Investigators are not necessarily police officers but can be civilians who are employed by the police force. They have a background in science; many have degrees in Forensic Science. The role of the crime scene investigator is to discover, record and preserve evidence.



What do crime scene investigators do?

When a serious crime takes place, the first thing the police do is to secure the crime scene. Everything is regarded as evidence so it is very important things remain untouched. Once the scene has been secured, then it is time to collect the evidence. This is where the crime scene investigator takes over. Their job is to visualise, capture and recover evidence using a range of techniques. This has to be carried out to agreed protocols so the evidence can be used in a court of law. Technology is constantly changing and the crime scene investigator has to know these new techniques. This is vitally important because as technology advances, crimes that could not be solved in the past now become solvable if the evidence has been preserved.

Detailed photographs and measurements of the crime scene are taken, including dusting for fingerprints, taking swabs for DNA analysis and looking for footprints.



Disclaimer

This resource is provided for informational and educational purposes only. As this resource refers to small items/loose parts which may present a choking risk, sharp equipment (cocktail sticks), plaster of Paris and putting paint on your feet, you must ensure that an adequate risk assessment is carried out prior to using this resource. Twinkl is not responsible for the health and safety of your group or environment. It is your responsibility to ensure the resource and the activity it contains are safe and appropriate to use in your situation. Supervising adults should check for allergens and assess any potential risks before the activities and only proceed if it is safe to do so.

Types of Evidence

Fingerprints

Fingerprints are unique; no two people have the same fingerprints, even identical twins. They also remain unchanged throughout our lives, unless the skin is damaged, in which case a new unique fingerprint will be formed. When we touch something, the oil and sweat on our fingers leaves an impression behind that can often be invisible.

Henry Faulds is credited as being the first person to realise that fingerprints could be used to identify and capture criminals. He first became interested in fingerprints when he noticed those left by the makers of ancient pots found at an archaeological site. In 1880, he published an article suggesting that fingerprints could be used to identify criminals, however, he did not receive proper recognition for his discovery during his lifetime.

It would not be until 1901 that fingerprint evidence would be accepted in law courts in the UK.

Fingerprints are grouped into three different types:

		
Loops	Whorls	Arches
These prints curve back on themselves to form a loop shape	These form circular or spiral patterns	These make a wave-like pattern across the finger



Loops make up around 65% of fingerprints, whorls about 30% and arches 5%.

Fingerprints Activity

Why not collect fingerprints from your friends and family?

You will need:

- An ink pad
- Squares marked on paper (one for each fingerprint)

Method:

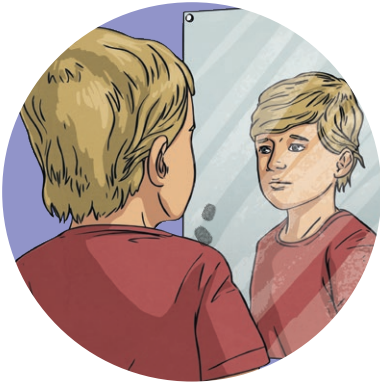
- 1 Press one fingertip onto an ink pad.
- 2 Press the fingertip onto one of the squares, careful not to smudge the print.
- 3 Repeat with the remaining fingers and thumbs.
- 4 Label each fingerprint loop, whorl or arch using the pictures from the table above.
- 5 Record whose prints they are and from which hand.

Name: _____				
Left/Right Hand				

Name: _____				
Left/Right Hand				

Can you find any fingerprints in your house?

Fingerprints are easiest to find on smooth surfaces, such as on windows, mirrors, tables or cutlery. Rough or porous surfaces do not preserve fingerprints very well. If you cannot find any fingerprints, you can always create some on a window or mirror.



You will need:

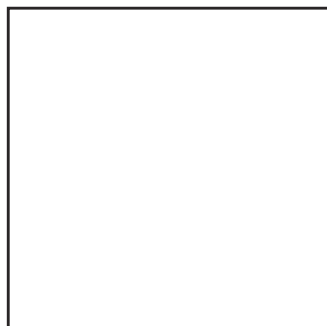
- Baby powder or cacao powder
- Soft paint brush or old makeup brush
- Clear sticky tape
- Hand lotion or cooking oil
- Magnifying glass

Method:

- 1** Using the brush and powder. Lightly dust the surface that you suspect of having fingerprints.
- 2** If you detect a fingerprint, press the clear, sticky tape over the print then gently lift away - careful not to smudge the print. The print will transfer onto the sticky tape.
- 3** Stick the tape onto the spaces provided below.
 - If you have used a light coloured powder use the black square.
 - If you've used a dark coloured powder use a light square.
- 4** Use your magnifying glass to look carefully at the print.
Can you identify whose fingerprint it is from the prints you collected in the first activity?

Note: If you have trouble getting a clear print, try rubbing a small amount of hand lotion or oil into your fingers before leaving a print on a window or mirror. This should leave a nice clear print.

- 5** Finish this exercise by cleaning any of the surfaces you dusted with powder.



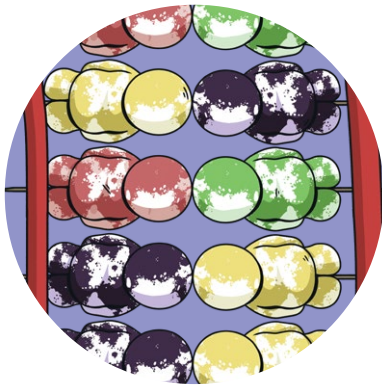
DNA

What is DNA?

DNA (short for deoxyribonucleic acid) is the single most important molecule in living cells. It contains all the information a person's body needs to grow, reproduce and function. You have about 9 million kilometres of DNA in your body. DNA is made up of two long strands that form a helix shape. The helix contains base pairs. It is the order of some sections of these base pairs that is unique and can be used to identify people. DNA carries our genetic code.

DNA Experiment

Try this fun, edible DNA experiment in order to better understand the structure of DNA. You will be making an edible model of the DNA double helix structure!

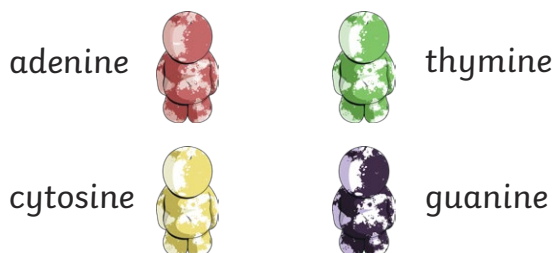


You will need:

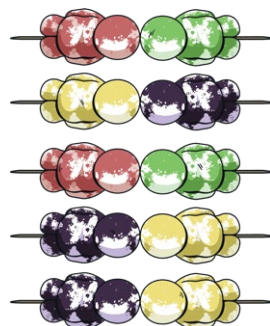
- Four different colours of soft sweets
- Two long cord sweets
- Cocktail sticks

Method:

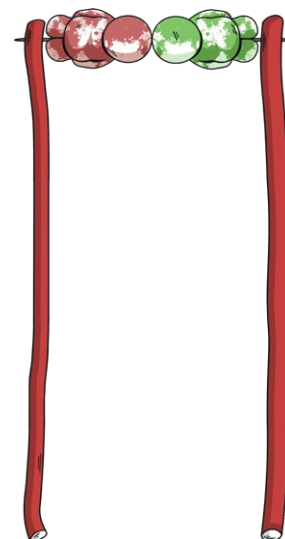
- 1 Choose a different colour of sweet to represent each base.



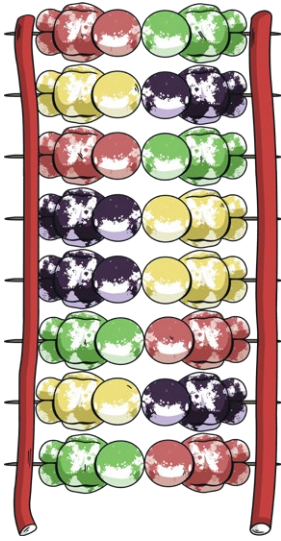
- 2 Thread two different coloured sweets onto a cocktail stick, making sure they form the correct base pairs.



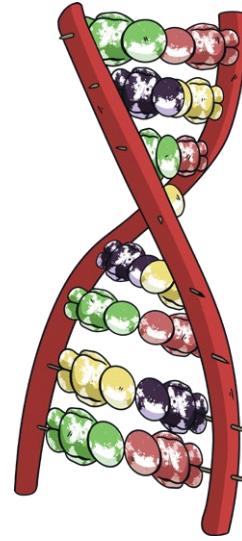
- 3 The cord sweets represent the sugar-phosphate backbone. Place one cord sweet at each end of one of the cocktail sticks.



- 4** Continue to join the cord sweets together with the cocktail sticks of base pairs.



- 5** Carefully hold each end of the structure and twist it slightly to represent the double helix.



So how does DNA help police solve crimes?

Alec Jeffreys, a genetics professor at Leicester University, had discovered in the 1980s that some patterns in DNA are unique to individuals. If DNA can be found at a crime scene, then that DNA can then be matched to a specific person and prove they were there.

What is the DNA database?

In the UK, the police take a sample of DNA from everyone who is arrested. This is done by swabbing the inside of the cheek. This DNA, known as profile records, is added to a database which was established in 1995. DNA samples taken from crime scenes can then be checked against this database. Between April 2001 and March 2016, over 610,000 unsolved crimes found matches to the database from DNA evidence that had been preserved from the crime scene.



Forensic Podiatry

A podiatrist is a specialist in treating problems with feet. Forensic podiatry is when a podiatrist uses their specialist knowledge of feet to help the police identify people. They can do this by analysing prints from shoes, the wear pattern on shoes and bare footprints.

They can also compare how someone walks, known as their gait. Suspects are often captured on CCTV but may be unidentifiable due to hoods or disguises. Podiatrists can look for distinctive features in the culprit's gait and compare it to the shoe patterns of the prints left and also the other shoes of the culprit not worn at the scene. Forensic podiatry can also be used to identify the height, weight and the sex of a person.

Detectives will often take a photograph or cast of a footprint for forensic podiatrists to examine.

Make your own footprint cast (WITH ADULT SUPERVISION ONLY)

Have a go at making a cast of your own footprint. This activity involves making a footprint in sand and taking a cast from that impression.

BEWARE! Do not put your foot directly into the plaster of Paris as this can result in severe burns.

You will need:

- A large container, such as a cardboard box or washing up bowl
- Sand
- Plaster of Paris
- Water
- Disposable plastic tub, such as a large margarine or ice cream container
- A stick or old spoon for stirring the plaster of Paris

Method:

- 1 Place the sand inside the large container, add water until it is slightly damp - this will help the sand retain its shape.
- 2 Create a footprint by pressing your foot firmly, with a steady even pressure, into the damp sand.
 - Take care when pulling your foot back, after making the footprint, not to disturb the impression you have made.
- 3 Mix the plaster of Paris according to the instructions on the packet.
 - **Take particular care to follow any safety instructions and do not allow contact with the skin. Use a disposable plastic tub.**

- 4 Pour the plaster of Paris slowly into the footprint. Pouring it slowly prevents too many air bubbles from forming.
- 5 Leave the plaster of Paris to set overnight.
- 6 Once set, remove the cast from the sand.

Footprint and Gait Analysis

This fun activity is great to do outside on a sunny day. The more people doing the activity, the more data you can collect.

You will need:

- Approximately two metres of lining paper. Lining paper is sold alongside wallpaper. You could also use wallpaper or sheets of newspaper taped together.
- Poster paint
 - If more than one person is doing the activity, you may like to have a different colour for each person.
- A tray with sides
- A washing up bowl
- Hand soap
- Water
- An old towel



Method:

- 1 Unroll the wallpaper, secure it if it curls up with tins or other heavy objects.
- 2 Pour a thin layer of poster paint onto the tray and place the tray at one end of the wallpaper.
 - Do not have the paint too thick or you will not leave good footprints.
- 3 Fill the washing up bowl with warm water and place it with the soap and towel at the opposite end of the wallpaper to the paint tray.
- 4 Step **carefully** into the paint, then walk along the wallpaper using your normal stride. You will leave footprints behind you on the wallpaper.
 - Try to keep walking and not look back until you reach the end. This will give you better prints.
- 5 Use the warm water, soap and towel to wash your feet straight away.
- 6 Change the paint colour for the next person but use the same wallpaper.



Footprint and Gait Analysis

There are various things you can record about feet and gait.

Measurements can be taken of the length of the foot and the length of the stride. This can give some indication of the size of the person. Feet also come in different shapes. Feet can have a low, medium or high arch. You can tell what sort of arch someone has by looking at their footprint as the footprint only shows the areas of the foot that came into contact with the ground.



LOW

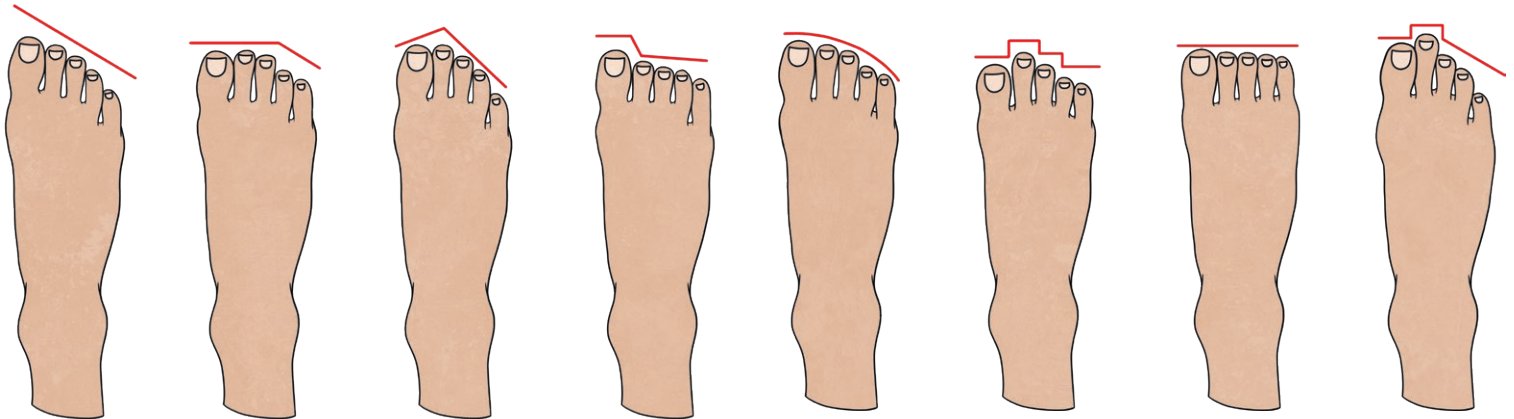


MEDIUM



HIGH

You can also look at the shape of the toes.



Results

Using the footprints you made on the wallpaper, complete the table for those who took part.

For the Distinguishing Features, observe whether the foot is arched - is it low, medium or high?
Can you describe the toes - which picture do they look like from those illustrated above?

	Footprints (name of person)	Length of Foot (cm)	Length of Stride (cm) - measure from heel to heel	Distinguishing Features
1				
2				
3				
4				
5				

Other Forms of Evidence

Finally, can you think of any other forms of evidence the police may use?

Not all evidence is taken from the crime scene. After a crime has been committed, police officers and detectives will carry out investigations in other areas. These might include:

- A. Social media - appeals for witnesses or people who passed through the area around the time
- B. Any video footage of the area from dashcams, door cameras and CCTV
- C. Interviews with people who knew the victim or anyone who was in the area at the time and may have seen anything to report
- D. Evidence from technology - mobile phones, cameras (police will review what is happening in the background of the photographs or video footage)

Using forensic evidence alongside these other forms, detectives will construct a timeline of events. The timeline is crucial if detectives are to understand what has happened and uncover the events leading up to the crime.

Look at this timeline and classify the different types of evidence mentioned. (Be reassured, the victim makes a full recovery!)

12:00	Suspect B's computer shows evidence of a Google Search: 'How to avoid being seen on CCTV'	
14:00	Suspect A phones the victim	
14:10	Suspect B sends Suspect A a message on social media	
15:00	Bank records show Suspect B withdrew £500 from a cash machine	
15:10	Dashcam on a bus shows Suspect A cycling near the bank where Suspect B withdrew the money	
17:00	Witnesses say Suspect B bought a weapon in a pub	
17:05	CCTV shows Suspect A and B leaving the pub	
17:30	Suspect B unfriends Suspect A on social media	
22:00	The injured body of the victim is found	